

All-Orders Asymptotic Transseries for the Butcher–Pólya Tree Numbers and Their Inverses

Rooted unlabeled trees, singularity sectors, Lambert inversion,
and coefficient formulae at arbitrary order

A self-contained research article prepared for the ProveIt project

Abstract. Let a_n be the number of rooted unlabeled non-plane trees with n vertices, the Butcher tree sequence OEIS A000081. Its ordinary generating function $T(z)$ satisfies Pólya’s equation

$$T(z) = z \exp \left(\sum_{k \geq 1} \frac{T(z^k)}{k} \right).$$

Writing $T(z) = C(\zeta(z))$, where $C(u) = -W_0(-u)$ is the Cayley tree function, separates a universal square-root singularity from an analytic problem-specific perturbation. We derive: (i) a Lagrange–Bürmann formula for every coefficient of the universal Cayley Puiseux expansion; (ii) an ordinary-Bell-polynomial formula for every local coefficient of T at its dominant singularity; (iii) a Bernoulli-polynomial formula for every term of the coefficient asymptotic expansion; and (iv) a recursive generalized-Bell calculus for every exponentially small singularity sector generated by analytic continuation. Numerically,

$$a_n \sim \frac{\rho^{-n}}{\sqrt{\pi n^3}} \left(0.77974501018732044277 + \frac{0.07828911261061096206}{n} + \frac{0.39294026766318601847}{n^2} + \dots \right),$$

with $\rho = 0.338321856899207695196112625717017 \dots$. For the asymptotic inverse, the leading exponential–power model is inverted exactly by the lower real branch W_{-1} , and all corrections are generated by one parameter-dependent Lagrange formula. This yields both a compact Lambert- W representation and the complete polynomial–logarithmic expansion with general coefficient formulae. For completeness, we also give the multiplicative reciprocal transseries and distinguish it from the local compositional inverse of the generating function.

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1 Introduction and scope

The numbers

$$0, 1, 1, 2, 4, 9, 20, 48, 115, 286, 719, 1842, 4766, 12486, 32973, 87811, \dots$$

form OEIS A000081. For $n \geq 1$, a_n counts isomorphism classes of rooted, unlabeled, non-plane trees on n vertices. They are often called *Pólya trees*; in numerical analysis the same rooted trees index Butcher series, elementary differentials, and Runge–Kutta order conditions, which motivates the name *Butcher tree sequence*.

The leading estimate

$$\begin{aligned} a_n &\sim A\lambda^n n^{-3/2}, \\ \lambda = \rho^{-1} &= 2.9557652856519949747\dots, \\ A &= 0.4399240125710253040\dots \end{aligned}$$

goes back to Pólya and Otter. Genitrini gave an explicit all-orders Puiseux and coefficient analysis for Pólya structures. The present article reorganizes that dominant calculation into a compact coefficient calculus and then develops two additional layers needed for the requested transseries and inverse:

1. a singularity-by-singularity, exponentially completed transseries, with a recursive formula for every local sector;
2. a full asymptotic inversion theory, first in a Lambert- W normal form and then in the polynomial–logarithmic scale used in the linked series-and-transseries notes.

1.1 What “full transseries” means here

There are two mathematically distinct senses of completeness.

Definition 1.1 (Dominant all-orders expansion). A *dominant all-orders expansion* is a Poincaré expansion at the unique singularity $z = \rho$ on the circle of convergence. For every fixed M it gives a remainder smaller by $M + 1$ algebraic powers of n .

Definition 1.2 (Exponentially completed expansion). An *exponentially completed expansion* augments the dominant sector by the Darboux sectors of singularities reached on analytically continued sheets. Each sector has its own exponential weight σ^{-n} and a full inverse power series. The unit circle is a natural boundary for the Pólya-tree generating function, so this completion is naturally a directed family over larger continuation domains, rather than one finite global sum.

The dominant sector developed below is rigorous and unconditional. The multi-sector formula is rigorous on any chosen continuation domain in which the relevant singularities are isolated algebraic singularities and a Cauchy contour can be deformed through their associated Hankel contours. Globally it is best understood as a formal resurgent/Darboux transseries whose sector multipliers depend on the continuation path and Stokes prescription.

1.2 Three meanings of inverse

For a sequence, the word *inverse* is ambiguous. We treat all three standard meanings and keep them notationally separate.

- The main object is the *asymptotic growth inverse*: solve $a_x = y$ for a large continuous index $x = x(y)$, then round to obtain a generalized inverse of the discrete sequence.
- The *multiplicative inverse* is the reciprocal $1/a_n$.

- The *generating-function inverse* is the formal compositional inverse $T^{(-1)}(w)$ near $w = 0$. Its coefficients form OEIS A050395 and are unrelated to inversion of the growth law.

The first is developed in full in section 10; the second in section 9; the third in section 13.

Principal result in normalized form. Set

$$\alpha = -\log \rho = 1.08375759724462466048271198089617\dots, \quad p = \frac{3}{2},$$

and

$$A = \frac{\kappa_0}{\sqrt{\pi}} = 0.439924012571025304040903391435\dots$$

Then

$$a_n \sim A e^{\alpha n} n^{-p} \left(1 + \sum_{m \geq 1} \frac{f_m}{n^m} \right),$$

$$f_m = \frac{\kappa_m}{\kappa_0},$$

where every κ_m is given explicitly by equation (40). If $X = X(y)$ denotes the exact Lambert core

$$X(y) = -\frac{p}{\alpha} W_{-1} \left(-\frac{\alpha}{p} \left(\frac{A}{y} \right)^{1/p} \right),$$

then the complete algebraic inverse is

$$x(y) \sim X(y) + \sum_{m \geq 1} \frac{d_m}{X(y)^m},$$

where every d_m is the finite coefficient extraction in equation (73). Expanding W_{-1} gives a complete log–log transseries with coefficients equation (84).

2 Coefficient calculus and conventions

The formulas are simplest with ordinary partial Bell coefficient polynomials.

Definition 2.1 (Ordinary partial Bell coefficient). For $n, k \geq 0$, define

$$\widehat{\mathcal{B}}_{n,k}(x_1, x_2, \dots) := [z^n] \left(\sum_{j \geq 1} x_j z^j \right)^k. \quad (1)$$

Thus $\widehat{\mathcal{B}}_{0,0} = 1$, $\widehat{\mathcal{B}}_{n,0} = 0$ for $n > 0$, and $\widehat{\mathcal{B}}_{n,k} = 0$ unless $k \leq n$.

The basic conversion rules are

$$[z^n] \left(1 + \sum_{j \geq 1} x_j z^j \right)^q = \sum_{k=0}^n \binom{q}{k} \widehat{\mathcal{B}}_{n,k}(x_1, x_2, \dots), \quad (2)$$

$$[z^n] \log \left(1 + \sum_{j \geq 1} x_j z^j \right) = \sum_{k=1}^n \frac{(-1)^{k-1}}{k} \widehat{\mathcal{B}}_{n,k}(x_1, x_2, \dots), \quad (3)$$

$$[z^n] \exp \left(\sum_{j \geq 1} x_j z^j \right) = \sum_{k=0}^n \frac{1}{k!} \widehat{\mathcal{B}}_{n,k}(x_1, x_2, \dots). \quad (4)$$

These are ordinary, not exponential, Bell polynomials. In terms of the standard exponential partial Bell polynomial $B_{n,k}$,

$$\widehat{\mathcal{B}}_{n,k}(x_1, x_2, \dots) = \frac{k!}{n!} B_{n,k}(1!x_1, 2!x_2, \dots).$$

We use $[z^n]F(z)$ for coefficient extraction, W_0 for the principal Lambert branch, W_{-1} for the lower real branch on $(-1/e, 0)$, and Log for a specified branch of the complex logarithm. Falling factorials are denoted $(x)_r = x(x-1) \cdots (x-r+1)$.

3 Butcher trees and the Pólya functional equation

Let

$$T(z) = \sum_{n \geq 1} a_n z^n, \quad a_1 = 1.$$

A rooted unlabeled tree is a root carrying an unlabeled multiset of rooted unlabeled trees. The cycle-index translation of this specification gives

$$T(z) = z \exp \left(\sum_{k \geq 1} \frac{T(z^k)}{k} \right). \quad (5)$$

Equivalently,

$$T(z) = z \prod_{m \geq 1} (1 - z^m)^{-a_m}. \quad (6)$$

3.1 Exact recurrence

Set

$$b_j = \sum_{d|j} d a_d. \quad (7)$$

Taking the logarithmic derivative of equation (6) and extracting coefficients gives the quadratic recurrence

$$(n-1)a_n = \sum_{j=1}^{n-1} b_j a_{n-j}, \quad n \geq 2. \quad (8)$$

Equivalently,

$$a_{n+1} = \frac{1}{n} \sum_{j=1}^n b_j a_{n-j+1}.$$

It computes $a_1, \dots, a_N$ in $\mathcal{O}(N^2)$ arithmetic operations and is the starting point for the high-precision constants later in the article.

3.2 Reduction to the universal Cayley function

Separate the $k = 1$ term in equation (5):

$$R(z) = \sum_{k \geq 2} \frac{T(z^k)}{k}, \quad \zeta(z) = z e^{R(z)}. \quad (9)$$

Then

$$T(z) = \zeta(z)e^{T(z)}. \quad (10)$$

Let $C(u)$ be the Cayley tree function, characterized by

$$C(u) = ue^{C(u)}, \quad C(0) = 0. \quad (11)$$

Hence

$$C(u) = -W_0(-u), \quad T(z) = C(\zeta(z)). \quad (12)$$

This factorization is the structural key: all square-root universality is in C , while all Pólya-tree-specific information is in the analytic map ζ .

4 The dominant singularity

Let ρ be the radius of convergence of T . Pólya's positivity and aperiodicity argument shows that ρ is the unique singularity on $|z| = \rho$. The implicit equation equation (10) can cease to be locally solvable only when

$$1 - \zeta(z)e^{T(z)} = 1 - T(z) = 0.$$

Consequently,

$$T(\rho) = 1, \quad \zeta(\rho) = e^{-1}. \quad (13)$$

Because $\rho < 1$, every z^k with $k \geq 2$ lies strictly inside the original disc when $|z| < \sqrt{\rho}$. Thus R and ζ are analytic in $|z| < \sqrt{\rho}$, and $\zeta'(\rho) > 0$. The generic quadratic branch point of C at $u = 1/e$ therefore transfers to a square-root singularity of T .

The defining scalar equation is

$$\log \rho + \sum_{k \geq 2} \frac{T(\rho^k)}{k} = -1. \quad (14)$$

Using equation (8) and rapidly convergent evaluations at ρ^k , $k \geq 2$, gives

$$\rho = 0.33832185689920769519611262571701705318377460753297 \dots, \quad (15)$$

$$\lambda = \rho^{-1} = 2.955765285651994974714817524123194588375 \dots, \quad (16)$$

$$\alpha = -\log \rho = 1.083757597244624660482711980896174279050 \dots \quad (17)$$

For later use,

$$\begin{aligned} h_1 &= e\rho\zeta'(\rho) = 1.21600456182404892215721261922818744 \dots, \\ \zeta'(\rho) &= 1.32224114269693026445128488981 \dots \end{aligned} \quad (18)$$

5 The universal Cayley Puiseux coefficients

The next theorem gives a compact all-order formula independent of the tree class.

Theorem 5.1 (Universal Cayley branch coefficients). *Put*

$$h = 1 - eu, \quad C(u) = 1 + \sum_{r \geq 1} c_r h^{r/2}. \quad (19)$$

Define

$$A(v) = \sum_{j \geq 0} \frac{2(j+1)}{(j+2)!} v^j = 1 + \frac{2}{3}v + \frac{1}{4}v^2 + \frac{1}{15}v^3 + \dots \quad (20)$$

Then, for every $r \geq 1$,

$$c_r = -\frac{2^{r/2}}{r} [v^{r-1}] A(v)^{-r/2}. \quad (21)$$

Equivalently,

$$c_r = -\frac{2^{r/2}}{r} \sum_{k=0}^{r-1} \binom{-r/2}{k} \widehat{\mathcal{B}}_{r-1,k} \left(\frac{2}{3}, \frac{1}{4}, \frac{1}{15}, \dots \right). \quad (22)$$

Proof. Set $q = 1 - C(u)$. From $u = Ce^{-C}$,

$$eu = (1 - q)e^q, \quad h = 1 - (1 - q)e^q = \sum_{m \geq 2} \frac{m-1}{m!} q^m = \frac{q^2}{2} A(q).$$

With $s = \sqrt{2h}$ this becomes $q = sA(q)^{-1/2}$. Lagrange–Bürmann inversion gives

$$[s^r]q = \frac{1}{r} [v^{r-1}] A(v)^{-r/2}.$$

Since $C = 1 - q$ and $s^r = 2^{r/2} h^{r/2}$, equation (21) follows. Expanding the generalized power by equation (2) gives equation (22). $\square$

The first values are

$$\begin{aligned} c_1 &= -\sqrt{2}, & c_2 &= \frac{2}{3}, & c_3 &= -\frac{11\sqrt{2}}{36}, & c_4 &= \frac{43}{135}, \\ c_5 &= -\frac{769\sqrt{2}}{4320}, & c_6 &= \frac{1768}{8505}, & c_7 &= -\frac{680863\sqrt{2}}{5443200}, & c_8 &= \frac{3926}{25515}. \end{aligned}$$

Formula equation (21) is often shorter than recursive descriptions of these coefficients and is well suited to symbolic computation.

6 Full local Puiseux expansion of the Pólya-tree function

Use the dimensionless local variable

$$X = 1 - \frac{z}{\rho}. \quad (23)$$

Define

$$h(X) = 1 - e\zeta(\rho(1 - X)) = \sum_{m \geq 1} h_m X^m. \quad (24)$$

The derivative form of the coefficients is

$$h_m = (-1)^{m+1} \frac{e\rho^m}{m!} \zeta^{(m)}(\rho). \quad (25)$$

In particular $h_1 > 0$.

6.1 A formula using only the tree numbers

For a fully constructive expression, write

$$\log(e\zeta(\rho(1 - X))) = \sum_{m \geq 1} \ell_m X^m. \quad (26)$$

Expanding $T((\rho(1 - X))^k)$ termwise gives

$$\ell_m = -\frac{1}{m} + (-1)^m \sum_{k \geq 2} \frac{1}{k} \sum_{r \geq 1} a_r \binom{kr}{m} \rho^{kr}. \quad (27)$$

The binomial coefficient is understood as zero for $kr < m$. Both sums converge rapidly because $\rho^k < \rho$ for $k \geq 2$. Since $h = 1 - \exp(\sum \ell_m X^m)$,

$$h_m = -\sum_{k=1}^m \frac{1}{k!} \widehat{\mathcal{B}}_{m,k}(\ell_1, \ell_2, \dots). \quad (28)$$

Thus the local singular coefficients are ultimately explicit in the exact integers a_r and the one scalar ρ . An equivalent derivative calculus follows from

$$L(z) = \log \zeta(z) = \log z + \sum_{k \geq 2} \frac{T(z^k)}{k},$$

for which

$$\zeta^{(m)}(\rho) = \zeta(\rho) Y_m(L'(\rho), L''(\rho), \dots, L^{(m)}(\rho)), \quad (29)$$

$$\frac{d^m}{dz^m} T(z^k) = \sum_{r=1}^m T^{(r)}(z^k) B_{m,r}((k)_1 z^{k-1}, \dots, (k)_{m-r+1} z^{k-m+r}), \quad (30)$$

$$T^{(r)}(\rho^k) = \sum_{n \geq r} (n)_r a_n \rho^{k(n-r)}. \quad (31)$$

Here Y_m and $B_{m,r}$ are the standard complete and partial exponential Bell polynomials. Equations (27) and (28) avoid this factorial rescaling and are usually preferable computationally.

6.2 General coefficient formula

Factor

$$h(X) = h_1 X \left(1 + \sum_{j \geq 1} q_j X^j \right), \quad q_j = \frac{h_{j+1}}{h_1}. \quad (32)$$

Since $T = C(\zeta)$, substitute equation (24) into the universal expansion equation (19).

Theorem 6.1 (All local Pólya-tree coefficients). *There is a convergent Puiseux expansion in a slit neighborhood of ρ ,*

$$T(z) = 1 + \sum_{n \geq 1} t_n X^{n/2}, \quad X = 1 - z/\rho. \quad (33)$$

For every $n \geq 1$,

$$t_n = \sum_{\substack{1 \leq r \leq n \\ r \equiv n \pmod{2}}} c_r h_1^{r/2} \sum_{k=0}^{(n-r)/2} \binom{r/2}{k} \widehat{\mathcal{B}}_{(n-r)/2,k}(q_1, q_2, \dots). \quad (34)$$

The c_r are given by equation (21); the h_j by equations (27) and (28).

Proof. For each r ,

$$h(X)^{r/2} = h_1^{r/2} X^{r/2} \left(1 + \sum_{j \geq 1} q_j X^j \right)^{r/2}.$$

To obtain $X^{n/2}$ one must have $n - r$ even, and the remaining coefficient is equation (2) with degree $(n - r)/2$. Summing over r gives equation (34). $\square$

The first analytic-input coefficients are

$$\begin{aligned} h_1 &= 1.2160045618240489221572\dots, & h_2 &= -0.4584157298563917191808\dots, \\ h_3 &= 0.4481047743895350436657\dots, & h_4 &= -0.3995616799562096774062\dots, \\ h_5 &= 0.3852513419466130007195\dots, & h_6 &= -0.3898065619592033419492\dots \end{aligned}$$

Selected singular coefficients are

$$\begin{aligned} t_1 &= -1.5594900203746408855422\dots, & t_3 &= -0.2854870216128456053039\dots, \\ t_5 &= -0.3424599704021542010422\dots, & t_7 &= -0.1077788002916310091822\dots, \\ t_9 &= -0.1952123835975564636128\dots, & t_{11} &= -0.0527247084981905623153\dots \end{aligned}$$

The even coefficients are part of the analytic germ at $z = \rho$. They do not contribute to the algebraic ρ^{-n} sector of $[z^n]T(z)$; after analytic continuation they are absorbed into more distant exponential sectors.

In Otter's coordinate $\rho - z = \rho X$, the leading coefficient is

$$T(z) = 1 - b(\rho - z)^{1/2} + \dots, \quad b = -\frac{t_1}{\sqrt{\rho}} = 2.6811281472671122386\dots$$

7 From the local expansion to the all-orders coefficient expansion

The exact binomial coefficient identity

$$[z^n](1 - z/\sigma)^\beta = \sigma^{-n} \frac{\Gamma(n - \beta)}{\Gamma(-\beta)\Gamma(n + 1)} \quad (35)$$

holds for $\beta \notin \{0, 1, 2, \dots\}$. It remains to expand the gamma ratio uniformly to arbitrary order.

7.1 Universal gamma-ratio coefficients

Let $B_m(x)$ denote the Bernoulli polynomial and define

$$\eta_r(\beta) = \frac{(-1)^{r+1}}{r(r+1)} (B_{r+1}(-\beta) - B_{r+1}(1)), \quad r \geq 1. \quad (36)$$

Define

$$\gamma_m(\beta) = [u^m] \exp \left(\sum_{r \geq 1} \eta_r(\beta) u^r \right) = \sum_{k=0}^m \frac{1}{k!} \widehat{\mathcal{B}}_{m,k}(\eta_1(\beta), \eta_2(\beta), \dots). \quad (37)$$

Then

$$\frac{\Gamma(n - \beta)}{\Gamma(n + 1)} \sim n^{-\beta-1} \sum_{m \geq 0} \frac{\gamma_m(\beta)}{n^m}. \quad (38)$$

For example,

$$\gamma_m\left(\frac{1}{2}\right) : \quad 1, \frac{3}{8}, \frac{25}{128}, \frac{105}{1024}, \frac{1659}{32768}, \frac{6237}{262144}, \dots$$

Theorem 7.1 (Dominant all-orders asymptotic expansion). *For every fixed $M \geq 0$,*

$$a_n = \frac{\rho^{-n}}{\sqrt{\pi} n^{3/2}} \left(\sum_{m=0}^M \frac{\kappa_m}{n^m} + \mathcal{O}(n^{-M-1}) \right), \quad n \rightarrow \infty, \quad (39)$$

where

$$\kappa_m = \sqrt{\pi} \sum_{j=0}^m \frac{t_{2j+1}}{\Gamma(-j - \frac{1}{2})} \gamma_{m-j} \left(j + \frac{1}{2} \right). \quad (40)$$

Every quantity on the right is given by a finite coefficient extraction in equations (21), (27), (28), (34) and (37).

Proof. Insert the odd part of equation (33) into Cauchy’s coefficient formula. For $t_{2j+1} X^{j+1/2}$, use equation (35) with $\beta = j + 1/2$, then equation (38). A contribution to the relative order n^{-m} has $j + r = m$, where r is the gamma-ratio order. This gives equation (40). Standard Δ -domain singularity analysis at the unique dominant singularity supplies the stated remainder. $\square$

The first eleven normalized coefficients are displayed in table 1. The first five agree with Genitrini’s independently computed values; the rest illustrate direct continuation of the same closed formula.

Table 1: Coefficients in $a_n \sim \rho^{-n} (\pi n^3)^{-1/2} \sum_{m \geq 0} \kappa_m n^{-m}$.

m	κ_m
0	0.7797450101873204427711
1	0.07828911261061096206128
2	0.3929402676631860184743
3	1.537879315978838091102
4	8.200844090435596159220
5	57.29291473494343787740
6	503.0445050262735818248
7	5359.600933884326033403
8	67342.06920114653048341
9	975425.4970695924732213
10	15996932.49293173312961

The leading constants are therefore

$$\kappa_0 = -\frac{t_1}{2} = 0.7797450101873204427711 \dots, \quad A = \frac{\kappa_0}{\sqrt{\pi}} = 0.4399240125710253040409 \dots \quad (41)$$

Writing

$$F(u) = 1 + \sum_{m \geq 1} f_m u^m, \quad f_m = \frac{\kappa_m}{\kappa_0}, \quad (42)$$

we obtain the inversion-ready normal form

$$a_n \sim A e^{\alpha n} n^{-p} F(1/n), \quad p = \frac{3}{2}. \quad (43)$$

The first normalized coefficients are

$$\begin{aligned} f_1 &= 0.10040348009640143476 \dots, & f_2 &= 0.50393431510230353310 \dots, \\ f_3 &= 1.9722849083822784692 \dots, & f_4 &= 10.517340904131573993 \dots, \\ f_5 &= 73.476474984020471064 \dots, & f_6 &= 645.13975524566129282 \dots \end{aligned}$$

Their factorial growth is expected: an all-orders singular expansion is normally asymptotic rather than convergent, and its late terms encode the next singularity actions.

8 The exponentially completed singularity transseries

The Pólya-tree function has one singularity on its initial circle $|z| = \rho$, but its analytic continuation has infinitely many singularities inside the unit disc, accumulating at every point of $|z| = 1$. This is a consequence of the substitutions $z \mapsto z^k$ in equation (5); the unit circle is a natural boundary. The dominant expansion in theorem 7.1 is unaffected, but a genuinely exponentially completed answer must retain the more distant singularities.

8.1 General local sector and coefficient transfer

Let σ be an isolated singularity on a chosen continuation sheet and write $X_\sigma = 1 - z/\sigma$. Suppose the local expansion has the generalized Puiseux form

$$T(z) = H_\sigma(z) + \sum_{\lambda \in \Lambda_\sigma} \tau_{\sigma,\lambda} X_\sigma^\lambda, \quad (44)$$

where H_σ is analytic, $\Lambda_\sigma \subset \mathbb{R}_{>0}$ is locally finite, and integer exponents are absorbed into H_σ . The associated all-orders Darboux sector is

$$\mathcal{A}_\sigma(n) \sim \sigma^{-n} \sum_{\lambda \in \Lambda_\sigma} \frac{\tau_{\sigma,\lambda}}{\Gamma(-\lambda)} n^{-\lambda-1} \sum_{m \geq 0} \frac{\gamma_m(\lambda)}{n^m}. \quad (45)$$

If $\bar{\sigma}$ is the conjugate singularity, the sum of the two sectors is real and oscillatory:

$$\mathcal{A}_\sigma(n) + \mathcal{A}_{\bar{\sigma}}(n) = 2|\sigma|^{-n} \Re \left(e^{-in \arg \sigma} \sum_{r \geq 0} c_{\sigma,r} n^{-\mu_\sigma - r} \right). \quad (46)$$

Relative to the dominant exponential ρ^{-n} , introduce the action

$$\Omega_\sigma = \text{Log}(\sigma/\rho). \quad (47)$$

Then $\sigma^{-n} = \rho^{-n} e^{-n\Omega_\sigma}$.

With Stokes multipliers S_σ fixed by a continuation prescription, the formal full transseries is

$$a_n \sim \mathcal{A}_\rho(n) + \sum_{\sigma \in \mathfrak{S} \setminus \{\rho\}} S_\sigma \mathcal{A}_\sigma(n), \quad (48)$$

where $\mathfrak{S}$ is the singular set encountered by the chosen contour continuation. Formula equation (45) supplies every coefficient once the local data are known.

8.2 A generalized Bell algebra for all local data

To compute the local data recursively, let Γ be the additive semigroup of exponents generated at a singularity and define

$$\mathfrak{B}_{\gamma,m}(x) = [X^\gamma] \left(\sum_{\lambda \in \Gamma_{>0}} x_\lambda X^\lambda \right)^m. \quad (49)$$

This is the ordinary Bell coefficient on a generalized power-series ring. All operations needed below are finite at a fixed exponent γ whenever Γ is well ordered.

Suppose $\sigma^k = \tau$ and near τ one has a term $t_{\tau,\lambda}(1 - w/\tau)^\lambda$. Since

$$1 - \frac{z^k}{\tau} = 1 - (1 - X_\sigma)^k = kX_\sigma \left(1 + \sum_{j=1}^{k-1} r_{k,j} X_\sigma^j \right), \quad r_{k,j} = \frac{(-1)^j}{k} \binom{k}{j+1}, \quad (50)$$

its pullback coefficient is explicitly

$$t_{\tau,\lambda} k^\lambda X_\sigma^\lambda \sum_{m \geq 0} \left[\sum_{q=0}^m \binom{\lambda}{q} \widehat{\mathcal{B}}_{m,q}(r_{k,1}, r_{k,2}, \dots) \right] X_\sigma^m. \quad (51)$$

Summing these pullbacks with the factors $1/k$ gives the local generalized series for $R(z)$.

If

$$R(z) = R_\sigma + \sum_{\lambda > 0} r_{\sigma,\lambda} X_\sigma^\lambda,$$

then

$$e^{R(z)-R_\sigma} = \sum_{\gamma \geq 0} E_{\sigma,\gamma} X_\sigma^\gamma, \quad E_{\sigma,\gamma} = \sum_{m \geq 0} \frac{1}{m!} \mathfrak{B}_{\gamma,m}(r_\sigma). \quad (52)$$

Multiplication by $z = \sigma(1 - X_\sigma)$ gives the local coefficients of $\zeta(z) = ze^{R(z)}$.

There are then two outer-composition rules.

Regular Cayley image. If $\zeta_\sigma \neq 1/e$ on the chosen branch and $C_b(u) = -W_b(-u)$ is analytic at ζ_σ , then

$$\tau_{\sigma,\gamma} = \sum_{m \geq 1} \frac{C_b^{(m)}(\zeta_\sigma)}{m!} \mathfrak{B}_{\gamma,m}(\zeta - \zeta_\sigma). \quad (53)$$

Critical Cayley image. If $\zeta_\sigma = 1/e$, form $h_\sigma = 1 - e\zeta(z)$ and use the universal coefficients:

$$T(z) = 1 + \sum_{r \geq 1} c_r h_\sigma(z)^{r/2}. \quad (54)$$

Generalized Bell extraction of each fractional power gives every $\tau_{\sigma,\gamma}$. A critical outer composition may halve existing exponents; resonant collisions are handled automatically by the semigroup calculus.

Algorithm 8.1 (Recursive construction of all sectors). Start from the dominant germ at ρ given by equation (34). For each candidate point σ and every $k \geq 2$ such that σ^k is a known singularity, pull the corresponding local series back by equation (51). Add the pullbacks to obtain R , exponentiate with equation (52), multiply by z , and apply either equation (53) or equation (54). Finally transfer the noninteger local powers with equation (45). Repeat outward in modulus.

The substitution mechanism forces the root-lift skeleton

$$\mathfrak{S}_{\text{lift}} = \left\{ \rho^{1/m} e^{2\pi i j/m} : m \geq 1, 0 \leq j < m \right\}. \quad (55)$$

Its radii tend to 1 and its arguments become dense. Further characteristic points can occur when a continued ζ again reaches $1/e$; theorem 8.1 treats them on the same footing.

8.3 The first explicit non-dominant sector

The negative point

$$\sigma_- = -\sqrt{\rho} = -0.581654413633394253810057781645\dots \quad (56)$$

is accessible along the negative real axis. Only the $k = 2$ term in $R(z)$ reaches the dominant singularity directly. With $X_- = 1 - z/\sigma_-$,

$$1 - \frac{z^2}{\rho} = 1 - (1 - X_-)^2 = 2X_- - X_-^2.$$

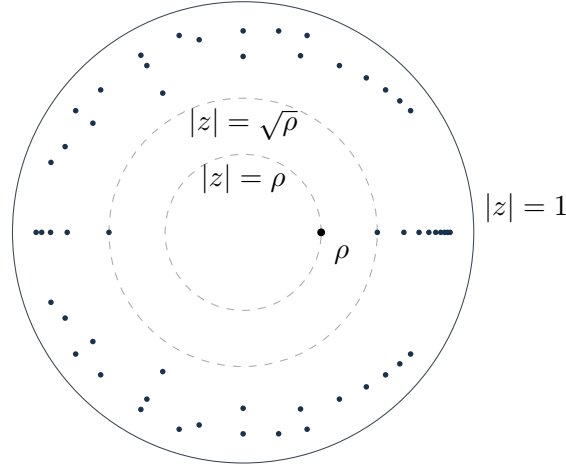


Figure 1: The first ten root-lift layers $z^m = \rho$. These are a structural skeleton of the analytically continued singularity set and accumulate at the natural boundary.

Therefore the leading fractional coefficient in R is $t_1/\sqrt{2}$. The regular Cayley outer map gives

$$\tau_{-,1/2} = \frac{T_-}{1 - T_-} \frac{t_1}{\sqrt{2}}, \quad T_- = C(\zeta_-), \quad \zeta_- = \sigma_- \exp\left(\frac{1}{2} + \sum_{k \geq 3} \frac{T(\sigma_-^k)}{k}\right). \quad (57)$$

Numerically,

$$\begin{aligned} \zeta_- &= -0.9293757352012050888501367\dots, \\ T_- &= -0.5410329414204534926205031\dots, \\ \tau_{-,1/2} &= 0.3871501110302429034772067\dots \end{aligned}$$

Thus the leading formal Darboux contribution is

$$\mathcal{A}_-(n) \sim -0.1092130299563101757054885\dots (-\sqrt{\rho})^{-n} n^{-3/2} (1 + \mathcal{O}(n^{-1})). \quad (58)$$

Relative to the dominant leading term, it begins as

$$\frac{\mathcal{A}_-(n)}{A\rho^{-n}n^{-3/2}} \sim -0.2482543049151649529915710\dots (-\sqrt{\rho})^n. \quad (59)$$

Its action is

$$\Omega_- = \frac{\alpha}{2} + i\pi \pmod{2\pi i}, \quad (60)$$

which explains the alternating exponentially small scale $(-1)^n \rho^{n/2}$ and the late-term divergence of the dominant inverse-power series.

9 Multiplicative reciprocal transseries

If “inverse” is intended as the reciprocal, the dominant algebraic answer is immediate from equation (43). Let

$$\frac{1}{F(u)} = 1 + \sum_{m \geq 1} r_m u^m. \quad (61)$$

Then

$$r_m = \sum_{k=1}^m (-1)^k \widehat{\mathcal{B}}_{m,k}(f_1, f_2, \dots), \quad m \geq 1. \quad (62)$$

Consequently,

$$\boxed{\frac{1}{a_n} \sim \frac{1}{A} e^{-\alpha n} n^{3/2} \left(1 + \sum_{m \geq 1} \frac{r_m}{n^m} \right)}. \quad (63)$$

The first coefficients are

$$\begin{aligned} r_1 &= -0.10040348009640143476 \dots, & r_2 &= -0.49385345628683505404 \dots, \\ r_3 &= -1.8721035437371763481 \dots, & r_4 &= -9.8824812214415511709 \dots, \\ r_5 &= -69.510824913364618832 \dots, & r_6 &= -616.91686459784338163 \dots \end{aligned}$$

For the full multisector reciprocal, write $a_n = M_n(1 + E_n)$, where M_n is the dominant all-orders sector and E_n is the sum of relative exponential sectors. Then

$$\frac{1}{a_n} = \frac{1}{M_n} \sum_{q \geq 0} (-E_n)^q. \quad (64)$$

Products generate all additive combinations of actions $\Omega_{\sigma_1} + \dots + \Omega_{\sigma_q}$, with coefficients obtained by the same generalized Bell convolution equation (49).

10 Asymptotic inverse of the coefficient sequence

The integer sequence has $a_1 = a_2 = 1$ and is strictly increasing thereafter. For large y , one may define the generalized inverse

$$N(y) = \min\{n \geq 2 : a_n \geq y\}. \quad (65)$$

An asymptotic series describes a smooth inverse $x(y)$; the discrete answer is obtained by rounding and can never be resolved more accurately than one unit without using the exact sequence. We invert the formal interpolation obtained from equation (43).

10.1 Exact Lambert core

First ignore $F(1/x)$ and solve

$$y = Ae^{\alpha x} x^{-p}, \quad p = \frac{3}{2}. \quad (66)$$

Taking the p th root and rearranging gives

$$-\frac{\alpha x}{p} \exp\left(-\frac{\alpha x}{p}\right) = -\frac{\alpha}{p} \left(\frac{A}{y}\right)^{1/p}.$$

The large positive solution requires the lower real branch:

$$\boxed{X(y) = -\frac{p}{\alpha} W_{-1}\left(-\frac{\alpha}{p} \left(\frac{A}{y}\right)^{1/p}\right)}. \quad (67)$$

This exact normalization absorbs all iterated logarithms caused by the factor x^{-p} .

10.2 All algebraic corrections by parameterized Lagrange inversion

Write

$$G(u) = \log F(u) = \sum_{m \geq 1} g_m u^m. \quad (68)$$

By equation (3),

$$g_m = \sum_{k=1}^m \frac{(-1)^{k-1}}{k} \widehat{\mathcal{B}}_{m,k}(f_1, f_2, \dots). \quad (69)$$

Seek $x = X + \delta$ and put $u = 1/X$. Since X solves the leading model, substitution into the logarithm of equation (43) yields exactly

$$\alpha\delta - p \log(1 + u\delta) + G\left(\frac{u}{1 + u\delta}\right) = 0. \quad (70)$$

Equivalently,

$$\delta = \Phi(u, \delta), \quad \Phi(u, v) = \frac{p}{\alpha} \log(1 + uv) - \frac{1}{\alpha} G\left(\frac{u}{1 + uv}\right). \quad (71)$$

There is a unique formal solution $\delta(u) \in u\mathbb{R}[[u]]$.

Theorem 10.1 (General coefficient formula for the inverse). *The complete algebraic inverse expansion is*

$$x(y) \sim X(y) + \sum_{m \geq 1} \frac{d_m}{X(y)^m}, \quad (72)$$

where

$$d_m = \sum_{k=1}^m \frac{1}{k} [u^m v^{k-1}] \Phi(u, v)^k. \quad (73)$$

The function Φ is given by equation (71), and its coefficients are finite Bell-polynomial expressions in $f_1, \dots, f_m$.

Proof. Treat u as a parameter in the formal fixed-point equation $v = \Phi(u, v)$. The Lagrange–Bürmann formula gives

$$\delta(u) = \sum_{k \geq 1} \frac{1}{k} [v^{k-1}] \Phi(u, v)^k.$$

Because $\Phi(u, v) = \mathcal{O}(u)$, only $k \leq m$ can contribute to $[u^m]$, which gives equation (73). Substitution into equation (70) proves formal inversion order by order. $\square$

The first symbolic coefficients are

$$d_1 = -\frac{g_1}{\alpha}, \quad (74)$$

$$d_2 = -\frac{\alpha g_2 + p g_1}{\alpha^2}, \quad (75)$$

$$d_3 = -\frac{\alpha^2 g_3 + \alpha g_1^2 + \alpha p g_2 + p^2 g_1}{\alpha^3}, \quad (76)$$

$$d_4 = -\frac{2\alpha^3 g_4 + 6\alpha^2 g_1 g_2 + 2\alpha^2 p g_3 + 5\alpha p g_1^2 + 2\alpha p^2 g_2 + 2p^3 g_1}{2\alpha^4}. \quad (77)$$

For the Butcher tree sequence,

$$g_1 = 0.10040348009640143476\dots, \quad g_2 = 0.49889388569456929357\dots,$$

$$g_3 = 1.9220255338418218216 \dots,$$

$$g_4 = 10.197396423373768871 \dots,$$

and

$$d_1 = -0.092643853525613129842 \dots,$$

$$d_2 = -0.58856303993134726281 \dots,$$

$$d_3 = -2.5966801674075310212 \dots,$$

$$d_4 = -13.149053399967065685 \dots,$$

$$d_5 = -85.498701514625566522 \dots,$$

$$d_6 = -711.26291483294656090 \dots,$$

$$d_7 = -7306.7882702849672566 \dots,$$

$$d_8 = -89561.760882637248600 \dots$$

As with the forward coefficients, this series is asymptotic and should be truncated near its least term.

10.3 Polynomial–logarithmic form and Lambert polynomials

To express the inverse without an unevaluated Lambert function, define

$$\varepsilon = \frac{\alpha}{p} \left(\frac{A}{y} \right)^{1/p}, \quad S = \log \frac{1}{\varepsilon} = \frac{1}{p} \log \frac{y}{A} + \log \frac{p}{\alpha}, \quad L = \log S. \quad (78)$$

The lower branch has the complete expansion

$$W_{-1}(-\varepsilon) = -S - L + \sum_{n \geq 1} \frac{(-1)^n P_n(L)}{S^n}. \quad (79)$$

The Lambert polynomials admit either of the equivalent general formulas

$$P_n(u) = \frac{(-1)^{n+1}}{(n+1)!} \left(\prod_{j=0}^{n-1} (D_u - j) \right) u^{n+1}, \quad (80)$$

$$= (-1)^{n+1} \sum_{k=0}^n \frac{s(n, k)}{(n+1-k)!} u^{n+1-k}, \quad (81)$$

where $s(n, k)$ are signed Stirling numbers of the first kind. In particular,

$$P_1(u) = u,$$

$$P_2(u) = \frac{u(u-2)}{2},$$

$$P_3(u) = \frac{u(2u^2 - 9u + 6)}{6},$$

$$P_4(u) = \frac{u(3u^3 - 22u^2 + 36u - 12)}{12}.$$

Set

$$V(v, L) = 1 + Lv - \sum_{n \geq 1} (-1)^n P_n(L) v^{n+1}. \quad (82)$$

Then $X = (p/\alpha)SV(S^{-1}, L)$. Substituting this into equation (72) yields

$$x(y) \sim \frac{p}{\alpha}(S + L) + \sum_{m \geq 1} \frac{H_m(L)}{S^m}. \quad (83)$$

Theorem 10.2 (General polynomial–log coefficient). *For every $m \geq 1$,*

$$H_m(L) = -\frac{p}{\alpha}(-1)^m P_m(L) + \sum_{j=1}^m d_j \left(\frac{\alpha}{p} \right)^j [v^{m-j}] V(v, L)^{-j}. \quad (84)$$

Thus H_m is an explicitly computable polynomial of degree m , and equation (84) together with equations (73) and (81) is a closed formula for every block of the inverse logarithmic transseries.

The first two blocks simplify to

$$H_1(L) = \frac{p}{\alpha}L - \frac{g_1}{p}, \quad (85)$$

$$H_2(L) = -\frac{p}{2\alpha}L^2 + \left(\frac{p}{\alpha} + \frac{g_1}{p}\right)L - \frac{g_1}{p} - \frac{\alpha g_2}{p^2}. \quad (86)$$

The hierarchy has the canonical form

$$x(y) = \frac{1}{\alpha} \log y + \frac{p}{\alpha} \log \log y + \text{constant} + \sum_{m \geq 1} \frac{\text{polynomial in } \log \log y}{(\log y)^m},$$

but equations (67) and (72) is numerically more stable because it resums all Lambert-generated log–log terms.

11 Exponential sectors under inversion

The inverse of the full forward transseries contains more than the algebraic corrections in equation (72). Write a continuous interpolation as

$$a(x) = M(x)(1 + E(x)), \quad (87)$$

where M is the complete dominant sector and E is the sum of relative exponential sectors. Let $x_0(y)$ be the formal inverse of M , given by equation (72), and put $x = x_0 + \eta$. Define

$$Q_x(v) = \frac{\log M(x+v) - \log M(x)}{v}, \quad Q_x(0) = (\log M)'(x). \quad (88)$$

The exact correction equation is

$$\eta = \Psi_x(\eta), \quad \Psi_x(v) = -\frac{\log(1 + E(x+v))}{Q_x(v)}. \quad (89)$$

A second Lagrange inversion gives

$$\boxed{\eta(x) = \sum_{k \geq 1} \frac{1}{k} [v^{k-1}] \Psi_x(v)^k.} \quad (90)$$

This single formula generates every one-sector and multi-sector correction, including all additive combinations of actions. Its linearization is

$$\eta(x) = -\frac{E(x)}{(\log M)'(x)} + \mathcal{O}(E(x)^2) \sim -\frac{E(x)}{\alpha}. \quad (91)$$

Suppose a forward relative sector has the form

$$E_\sigma(x) = e^{-\Omega_\sigma x} x^{\nu_\sigma} \mathcal{F}_\sigma(1/x). \quad (92)$$

Using $y \sim Ae^{\alpha x} x^{-p}$,

$$e^{-\Omega_\sigma x} \sim \left(\frac{y}{A}\right)^{-\Omega_\sigma/\alpha} x^{-p\Omega_\sigma/\alpha}. \quad (93)$$

Thus an exponentially small term in the index becomes a generally complex power of y multiplied by powers of $\log y$. Conjugate actions combine to produce log-periodic oscillations in the inverse variable.

For the first alternating action equation (60),

$$\frac{\Omega_-}{\alpha} = \frac{1}{2} + i\frac{\pi}{\alpha}, \quad (94)$$

so the inverse has corrections whose envelope is approximately $y^{-1/2}(\log y)^{-3/4}$ and whose phase is periodic in $\log y$ up to lower logarithmic corrections. Formula equation (90) supplies all coefficients once a Stokes prescription for the forward sector is fixed.

12 Numerical validation

All numbers in this section were computed from the exact recurrence equation (8), the scalar equation (14), and the coefficient formulas above using high-precision arithmetic. No fit to the sequence was used.

12.1 Forward expansion

Let $A_M(n)$ denote equation (39) truncated after κ_M/n^M . Table 2 reports the signed relative error $(A_M(n) - a_n)/a_n$. The improvement is characteristic of an asymptotic series: for fixed n the error first falls rapidly, then eventually rises if the series is over-truncated.

Table 2: Signed relative error of the forward approximation.

n	$M = 0$	$M = 2$	$M = 4$	$M = 6$	$M = 8$	$M = 10$
10	$-1.52 \cdot 10^{-2}$	$-3.49 \cdot 10^{-4}$	$2.63 \cdot 10^{-3}$	$3.99 \cdot 10^{-3}$	$5.52 \cdot 10^{-3}$	$8.77 \cdot 10^{-3}$
20	$-6.58 \cdot 10^{-3}$	$-3.42 \cdot 10^{-4}$	$-3.18 \cdot 10^{-5}$	$1.04 \cdot 10^{-6}$	$9.73 \cdot 10^{-6}$	$1.41 \cdot 10^{-5}$
50	$-2.22 \cdot 10^{-3}$	$-1.77 \cdot 10^{-5}$	$-2.88 \cdot 10^{-7}$	$-1.20 \cdot 10^{-8}$	$-9.84 \cdot 10^{-10}$	$-1.36 \cdot 10^{-10}$
100	$-1.06 \cdot 10^{-3}$	$-2.08 \cdot 10^{-6}$	$-8.06 \cdot 10^{-9}$	$-7.88 \cdot 10^{-11}$	$-1.50 \cdot 10^{-12}$	$-4.73 \cdot 10^{-14}$
200	$-5.15 \cdot 10^{-4}$	$-2.53 \cdot 10^{-7}$	$-2.40 \cdot 10^{-10}$	$-5.73 \cdot 10^{-13}$	$-2.66 \cdot 10^{-15}$	$-2.04 \cdot 10^{-17}$
500	$-2.03 \cdot 10^{-4}$	$-1.59 \cdot 10^{-8}$	$-2.39 \cdot 10^{-12}$	$-9.02 \cdot 10^{-16}$	$-6.62 \cdot 10^{-19}$	$-8.01 \cdot 10^{-22}$

12.2 Inverse expansion

For $y = a_n$, the exact continuous inverse should return n . Let X be the Lambert core equation (67), and let $x_M = X + \sum_{j=1}^M d_j X^{-j}$. Table 3 gives $x_M(a_n) - n$.

Table 3: Absolute index error of the Lambert-normalized inverse.

n	core X	$M = 1$	$M = 2$	$M = 4$	$M = 6$	$M = 8$
10	$1.64 \cdot 10^{-2}$	$7.15 \cdot 10^{-3}$	$1.29 \cdot 10^{-3}$	$-2.61 \cdot 10^{-3}$	$-4.16 \cdot 10^{-3}$	$-5.76 \cdot 10^{-3}$
20	$6.55 \cdot 10^{-3}$	$1.91 \cdot 10^{-3}$	$4.44 \cdot 10^{-4}$	$3.76 \cdot 10^{-5}$	$-2.02 \cdot 10^{-7}$	$-9.39 \cdot 10^{-6}$
50	$2.11 \cdot 10^{-3}$	$2.59 \cdot 10^{-4}$	$2.32 \cdot 10^{-5}$	$3.32 \cdot 10^{-7}$	$1.26 \cdot 10^{-8}$	$9.98 \cdot 10^{-10}$
100	$9.88 \cdot 10^{-4}$	$6.16 \cdot 10^{-5}$	$2.74 \cdot 10^{-6}$	$9.34 \cdot 10^{-9}$	$8.35 \cdot 10^{-11}$	$1.53 \cdot 10^{-12}$
200	$4.78 \cdot 10^{-4}$	$1.50 \cdot 10^{-5}$	$3.33 \cdot 10^{-7}$	$2.79 \cdot 10^{-10}$	$6.09 \cdot 10^{-13}$	$2.71 \cdot 10^{-15}$
500	$1.88 \cdot 10^{-4}$	$2.38 \cdot 10^{-6}$	$2.10 \cdot 10^{-8}$	$2.78 \cdot 10^{-12}$	$9.59 \cdot 10^{-16}$	$6.75 \cdot 10^{-19}$

The inverse error mirrors the forward error because local inversion divides a relative forward perturbation by approximately $(\log a)' = \alpha$. For practical inversion at moderate size, one should truncate near the least term and then round to the nearest admissible integer, checking adjacent values with the exact recurrence if a certified discrete answer is required.

13 Other inverse notions

13.1 The formal compositional inverse of the OGF

Because $T(z) = z + \mathcal{O}(z^2)$, it has a formal inverse

$$Z(w) = T^{(-1)}(w) = w - w^2 + w^4 - w^5 + w^6 - 4w^7 + 11w^8 - \dots \quad (95)$$

Lagrange inversion gives the exact coefficient formula

$$[w^n]Z(w) = \frac{1}{n}[z^{n-1}] \left(\frac{z}{T(z)} \right)^n = \frac{1}{n}[z^{n-1}] \exp \left(-n \sum_{k \geq 1} \frac{T(z^k)}{k} \right). \quad (96)$$

Together with equation (4), this is an explicit finite Bell-polynomial formula in $a_1, \dots, a_{n-1}$.

Near the dominant value $w = 1$, the inverse branch is actually analytic in $(1 - w)$ because it resolves the square root. From $T = \zeta e^T$,

$$Z(w) = \zeta^{\langle -1 \rangle} (w e^{-w}) \quad (97)$$

on the local branch through $Z(1) = \rho$. Since $w e^{-w} - e^{-1}$ begins quadratically at $w = 1$, the local expansion of Z has no linear term there. This formal inverse is conceptually different from the large- y index inverse developed in section 10.

13.2 Inverse branches of the Cayley factorization

Equation (97) is also useful for local coefficient calculus. If

$$\zeta(\rho + x) - e^{-1} = x\varphi(x), \quad \varphi(0) = \zeta'(\rho) \neq 0, \quad (98)$$

then

$$[v^m](\zeta^{\langle -1 \rangle}(e^{-1} + v) - \rho) = \frac{1}{m} [x^{m-1}] \varphi(x)^{-m}. \quad (99)$$

Composing with

$$(1 - s)e^{-(1-s)} - e^{-1} = -e^{-1} \sum_{m \geq 2} \frac{m-1}{m!} s^m$$

gives every Taylor coefficient of $Z(1 - s)$ by another finite ordinary-Bell convolution.

14 Reproducible coefficient pipeline

The following pipeline produces the forward, reciprocal, and inverse coefficients to any prescribed finite order.

Algorithm 14.1 (Dominant and inverse coefficients to order M). 1. Compute sufficiently many exact a_n using equations (7) and (8).

2. Solve equation (14) for ρ at the desired precision.
3. For $1 \leq m \leq M + 1$, compute ℓ_m from equation (27), then h_m from equation (28).
4. Compute c_r from equation (21) and t_r from equation (34) through $r = 2M + 1$.
5. Compute $\gamma_j(j' + 1/2)$ from equations (36) and (37), then κ_m from equation (40).
6. Normalize $f_m = \kappa_m / \kappa_0$. Obtain reciprocal coefficients r_m from equation (62).
7. Obtain g_m from equation (69) and inverse coefficients d_m from equation (73). If a pure log-log form is needed, generate P_m using equation (81) and H_m using equation (84).
8. For exponentially small sectors, apply theorem 8.1 in increasing modulus and transfer each local germ using equation (45). Invert the resulting transseries with equation (90).

At fixed M , every Bell sum is finite. The only infinite numerical sums are those in equations (14) and (27); they converge geometrically because their arguments are at most ρ^2 . Consequently, high-order coefficients can be generated with controlled arbitrary-precision arithmetic without numerical differentiation at a singular point.

15 Conclusions

The Pólya functional equation becomes particularly transparent after the factorization

$$T = C \circ \zeta, \quad C(u) = -W_0(-u).$$

The universal part is governed by one Lagrange coefficient $[v^{r-1}]A(v)^{-r/2}$; the problem-specific part is governed by ordinary Bell composition; and coefficient transfer is governed by Bernoulli polynomials in the gamma-ratio expansion. These three layers lead to the closed master formula equation (40) for every algebraic coefficient of A000081.

The same structure makes inversion systematic. The exponential–power core is inverted exactly with W_{-1} , the remaining algebraic corrections are a single parameterized Lagrange series equation (73), and the complete polynomial–logarithmic blocks follow from equation (84). More distant singularities produce exponentially small forward sectors and complex-power, log-periodic inverse sectors; their coefficients are generated recursively by the generalized Bell calculus of section 8.

The natural boundary prevents the global transseries from collapsing to a finite list of exponentials. It does not prevent an all-orders description: for each finite continuation domain, the local-sector recursion and transfer formula give the complete expansion associated with every singularity encountered there, and these finite descriptions form a coherent directed system as the domain approaches the unit circle.

A Additional low-order inverse coefficients

For reference, the fifth inverse coefficient is

$$d_5 = -\frac{1}{2\alpha^5} \left(2\alpha^4 g_5 + 8\alpha^3 g_1 g_3 + 4\alpha^3 g_2^2 + 2\alpha^3 p g_4 + 4\alpha^2 g_1^3 \right. \\ \left. + 14\alpha^2 p g_1 g_2 + 2\alpha^2 p^2 g_3 + 9\alpha p^2 g_1^2 + 2\alpha p^3 g_2 + 2p^4 g_1 \right).$$

Higher expressions grow rapidly in size; equation (73) is both shorter and less error-prone.

B Why analytic local terms do not enter the dominant sector

A nonnegative integer power $(1 - z/\rho)^m$ is a polynomial and has no coefficients past degree m . More generally, the entire analytic germ at $z = \rho$ can be continued through ρ and therefore contributes on scales determined by its next singularities, not on the algebraic $\rho^{-n}n^{-j}$ scale. This is why only the odd Puiseux coefficients t_{2j+1} appear in equation (40). In an exponentially completed analysis, the analytic germ is not discarded; it reappears through the sector recursion at more distant singularities.

C References

References

- [1] J. P. Bell, S. N. Burris, and K. A. Yeats, Counting rooted trees: the universal law $t(n) \sim C\rho^{-n}n^{-3/2}$, *Electronic Journal of Combinatorics* 13 (2006), R63, <https://doi.org/10.37236/1089>.
- [2] J. C. Butcher, *Numerical Methods for Ordinary Differential Equations*, third edition, Wiley, 2016.
- [3] L. Comtet, *Advanced Combinatorics: The Art of Finite and Infinite Expansions*, Reidel, 1974.

- [4] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, On the Lambert W function, *Advances in Computational Mathematics* 5 (1996), 329–359, <https://doi.org/10.1007/BF02124750>.
- [5] M. Drmota and B. Gittenberger, The shape of unlabeled rooted random trees, *European Journal of Combinatorics* 31 (2010), 2028–2063, <https://arxiv.org/abs/1003.1322>.
- [6] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009, <https://doi.org/10.1017/CBO9780511801655>.
- [7] A. Genitrini, Full asymptotic expansion for Pólya structures, in *Proceedings of the 27th International Conference on Probabilistic, Combinatorial and Asymptotic Methods for the Analysis of Algorithms*, 2016, <https://arxiv.org/abs/1605.00837>.
- [8] F. Harary, R. W. Robinson, and A. J. Schwenk, Twenty-step algorithm for determining the asymptotic number of trees of various species, *Journal of the Australian Mathematical Society, Series A* 20 (1975), 483–503.
- [9] D. E. Knuth, *The Art of Computer Programming, Volume 1: Fundamental Algorithms*, third edition, Addison–Wesley, 1997.
- [10] OEIS Foundation, A000081: number of unlabeled rooted trees with n nodes, <https://oeis.org/A000081>, accessed September 3, 2026.
- [11] R. Otter, The number of trees, *Annals of Mathematics* 49 (1948), 583–599, <https://doi.org/10.2307/1969046>.
- [12] G. Pólya, Kombinatorische Anzahlbestimmungen für Gruppen, Graphen und chemische Verbindungen, *Acta Mathematica* 68 (1937), 145–254, <https://doi.org/10.1007/BF02546665>.
- [13] V. Reshetnikov, Combinatorial Coefficient Calculus, ProveIt repository, https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/combinatorial-coefficient-calculus/Combinatorial_Coefficient_Calculus, accessed September 3, 2026.
- [14] V. Reshetnikov, Series and transseries research drafts, ProveIt repository, <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/series-and-transseries>, accessed September 3, 2026.